

Globular Cluster Intro Slides

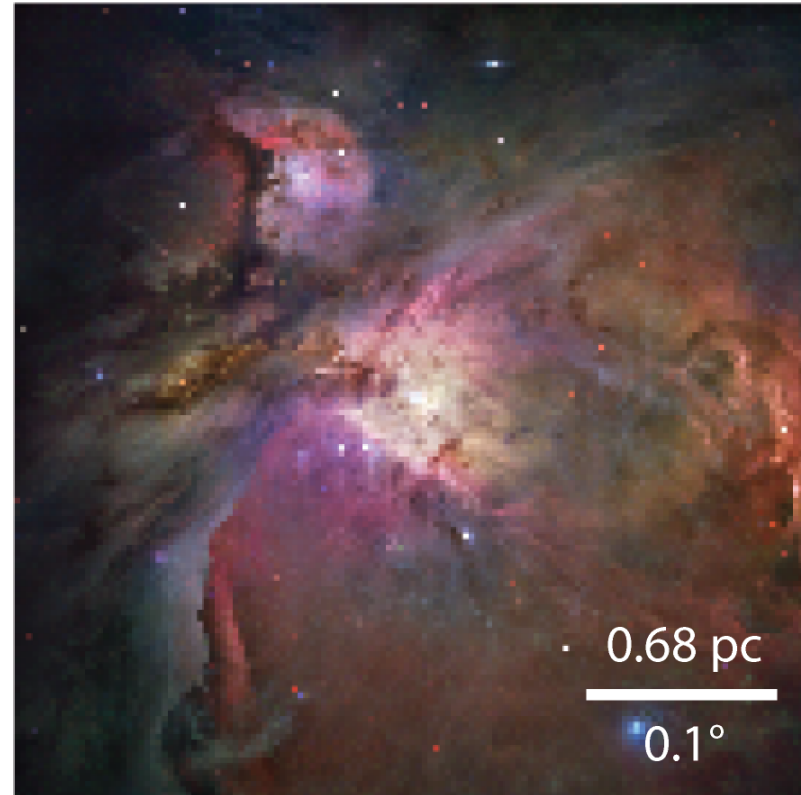
AY 250: Stellar Populations

Feb 24, 2026

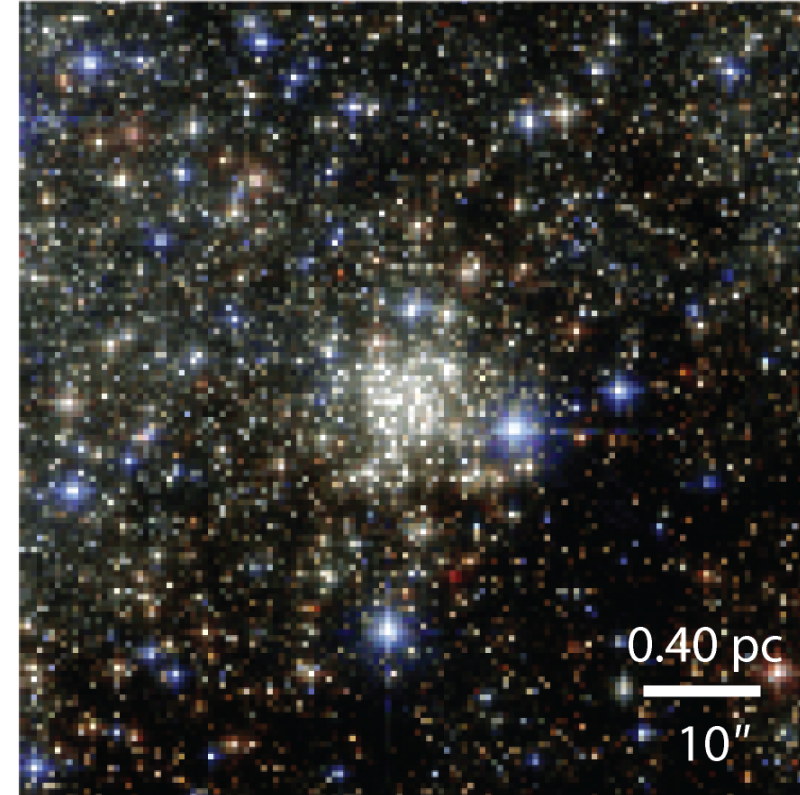
Dan Weisz

Stars Clusters of Different Ages

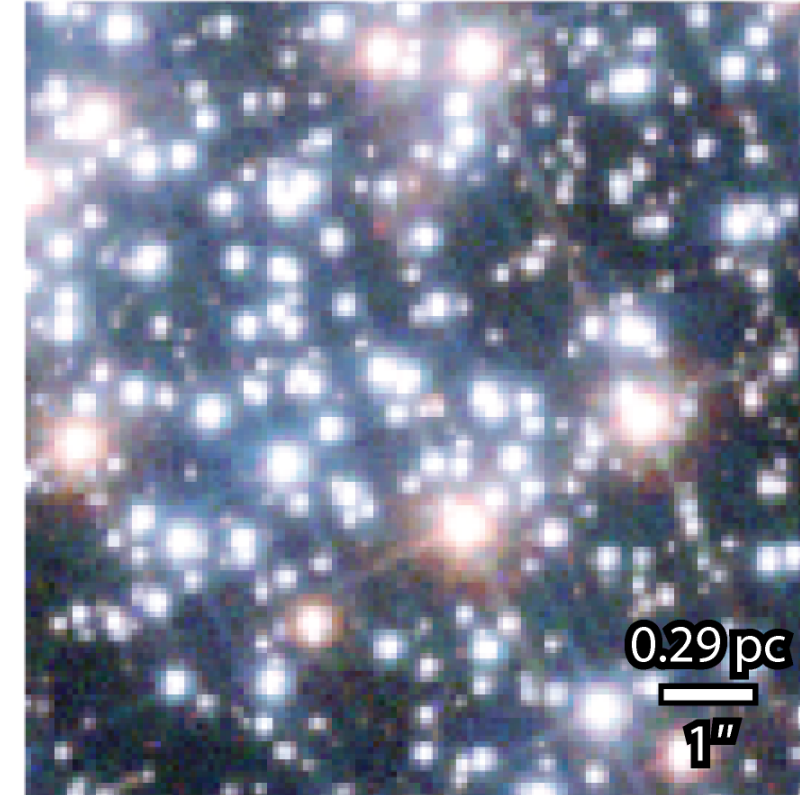
a ONC, $T < 3$ Myr



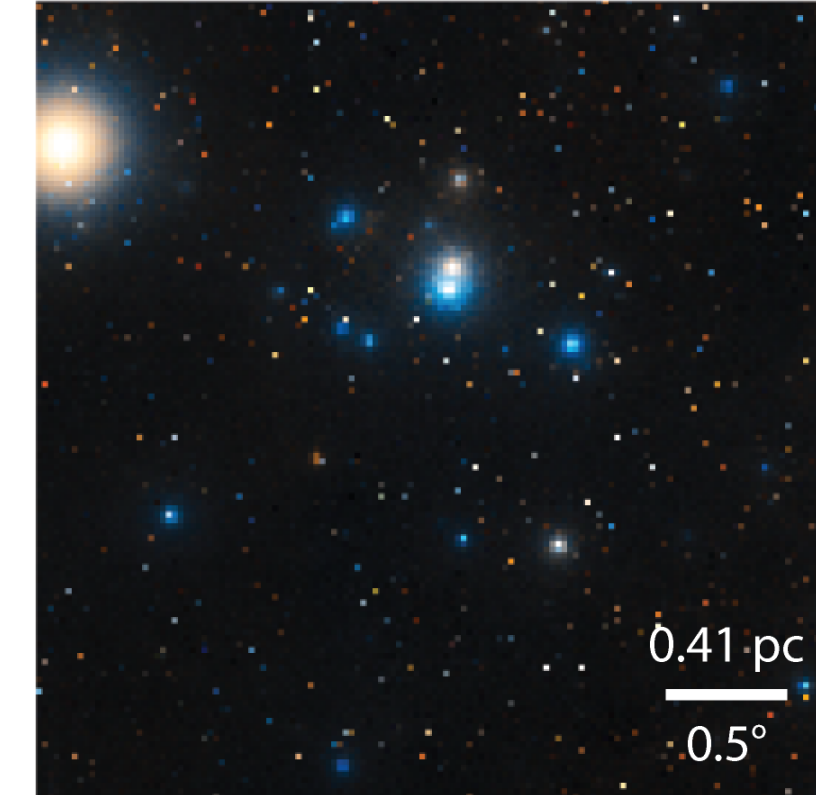
b Arches, $T \approx 2-3$ Myr



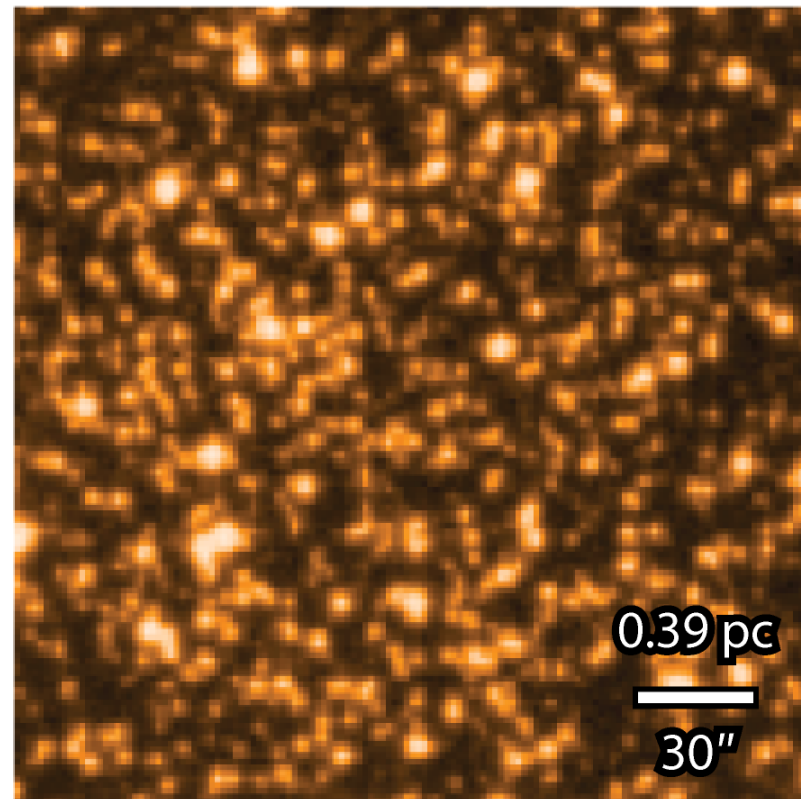
c NGC 265, $T \approx 250$ Myr



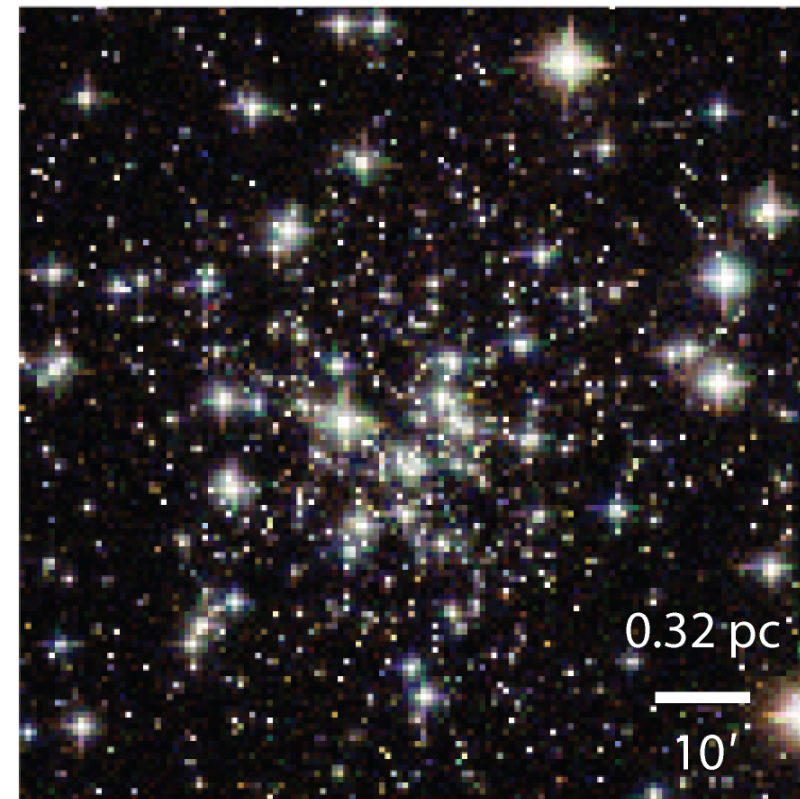
d Hyades, $T \approx 625$ Myr



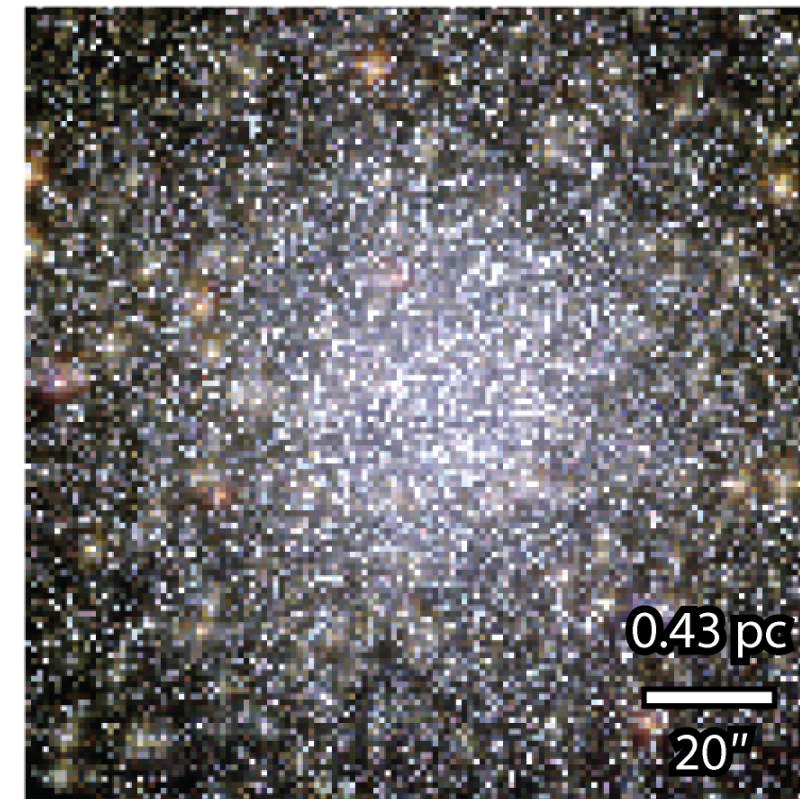
e Coll 261, $T \approx 7$ Gyr



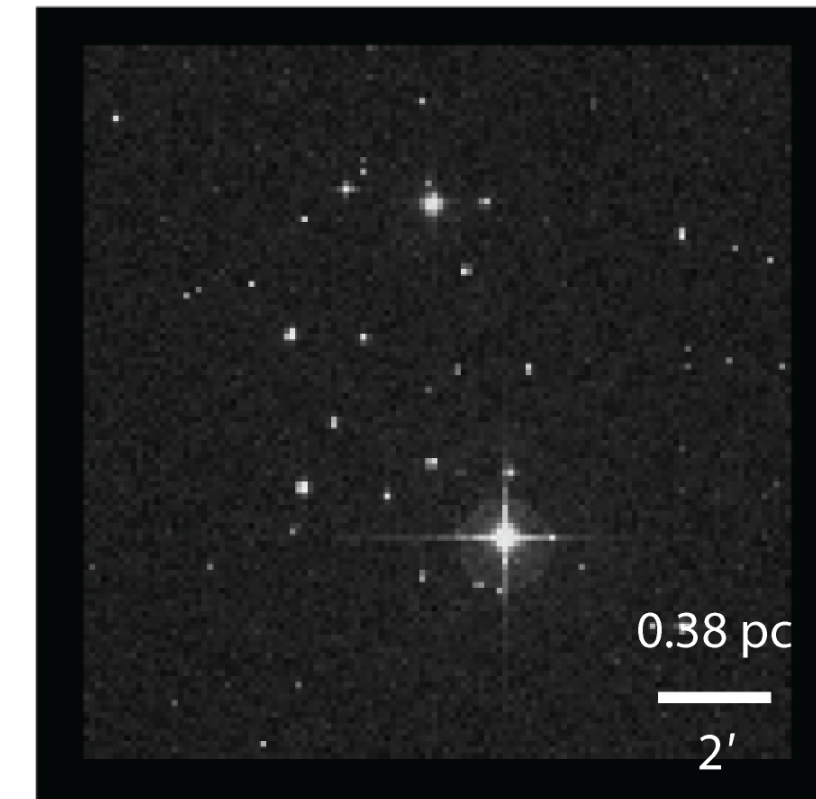
f NGC 6535, $T > 10$ Gyr



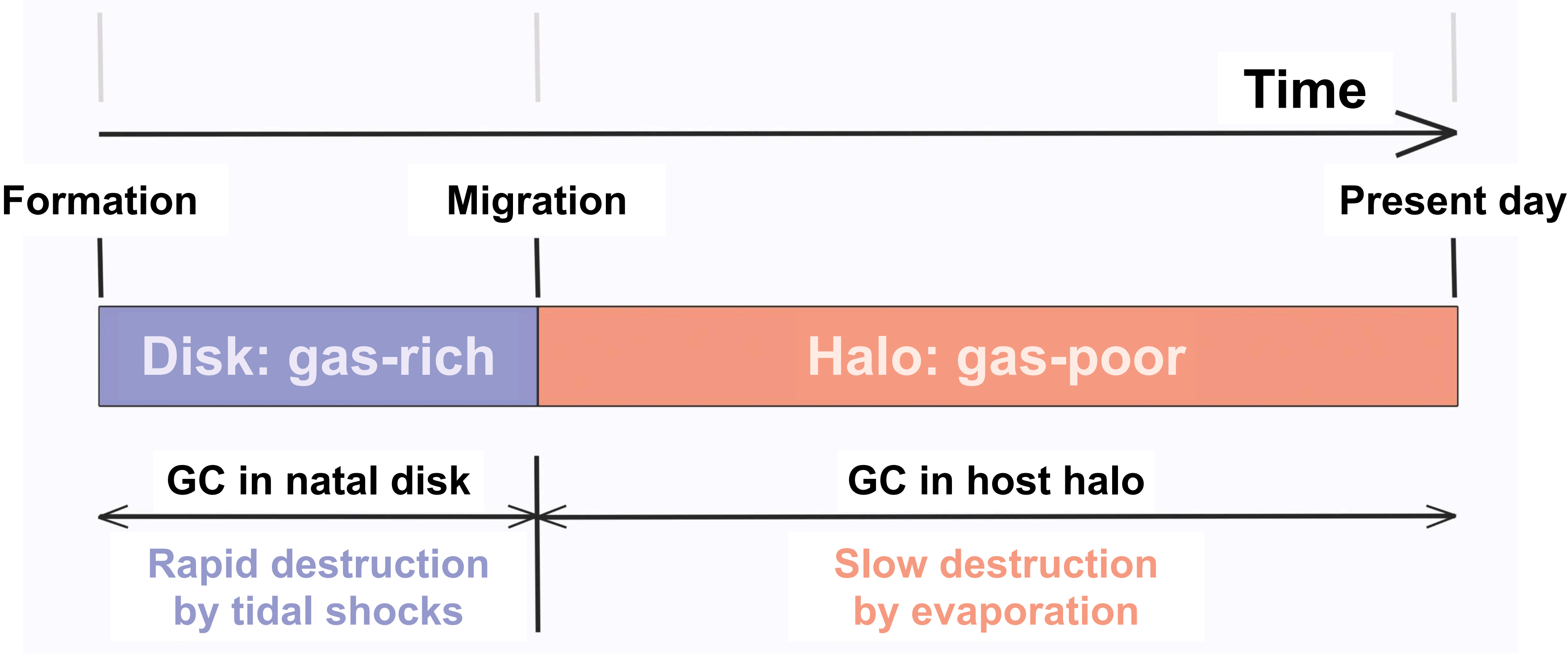
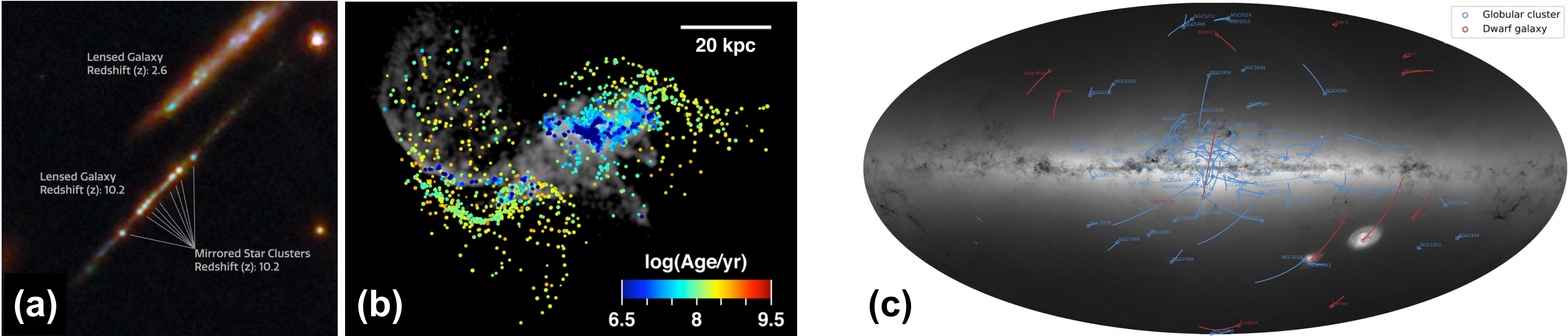
g 47 Tuc, $T > 10$ Gyr



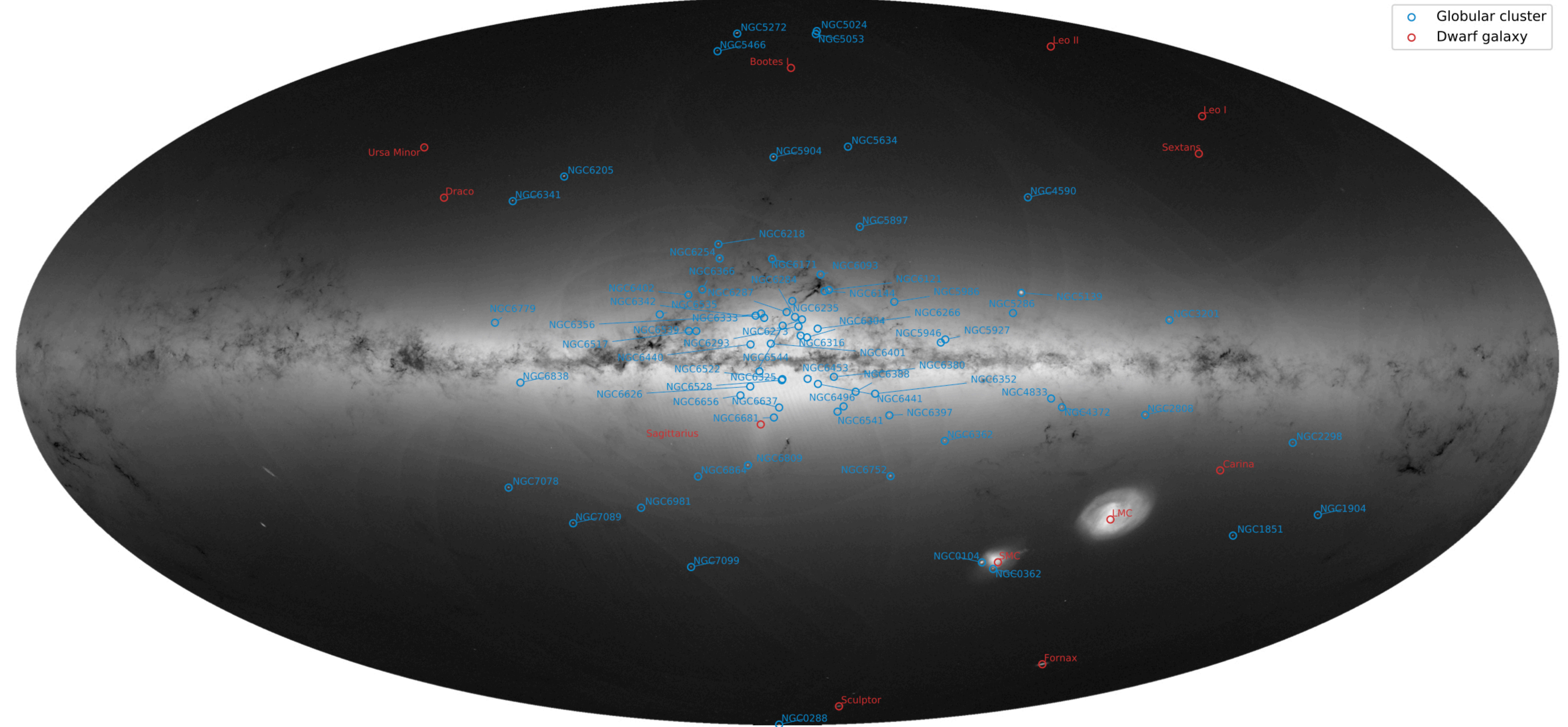
h NGC 1252: Fake News



One Model for Globular Cluster Formation in a MW-like Galaxy



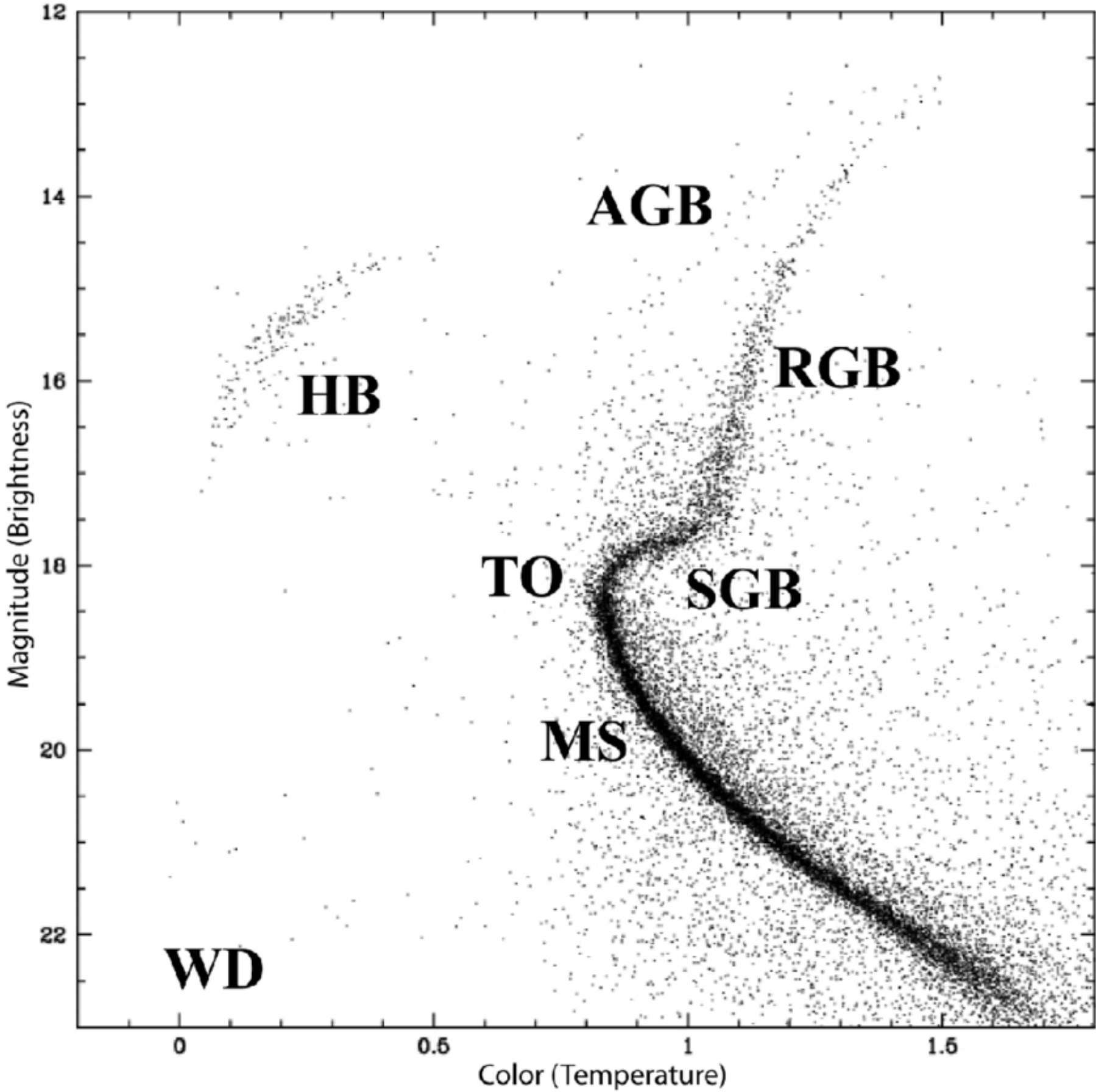
Gaia-based Demographics of Milky Way Globular Clusters



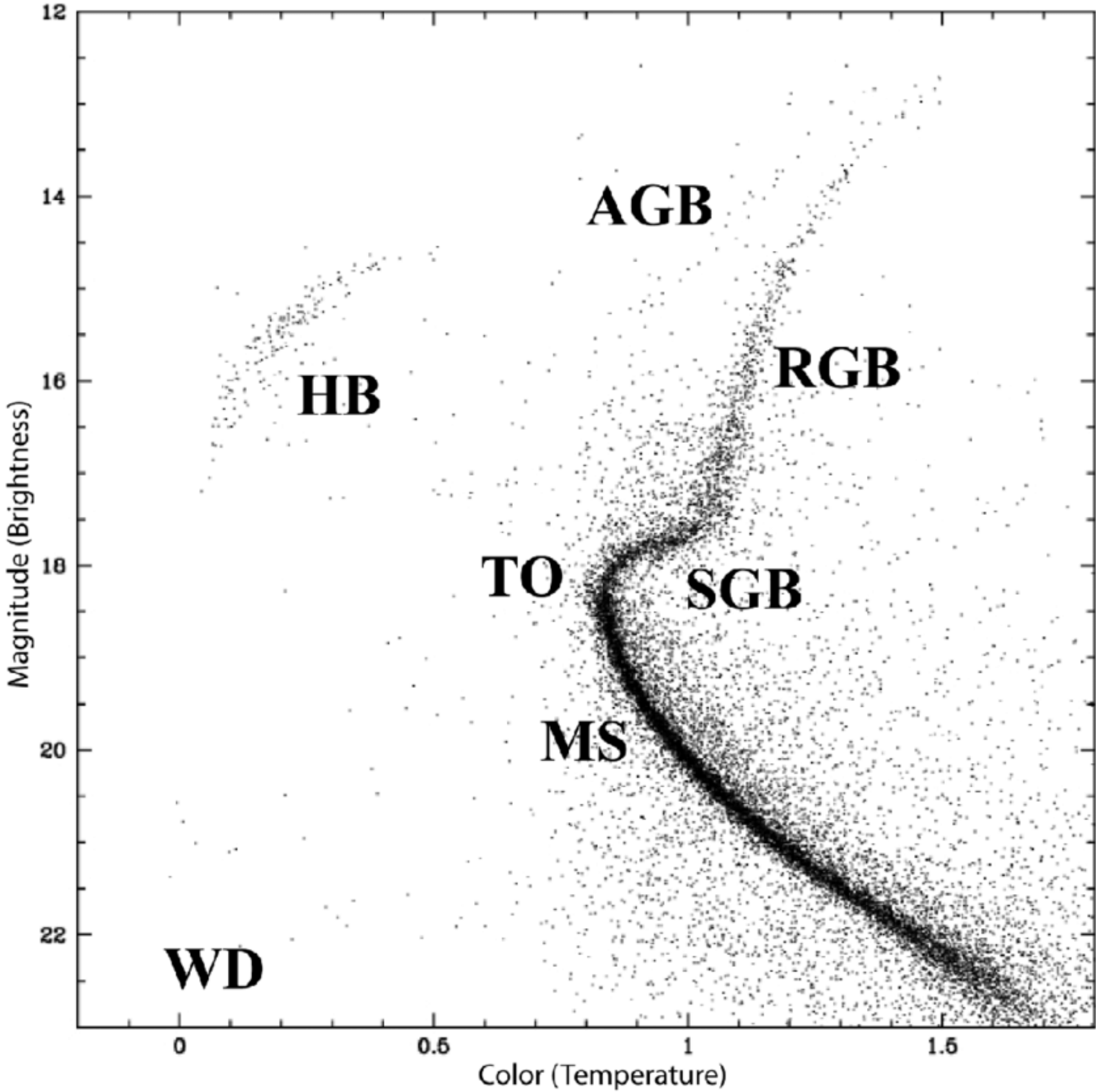
https://www.esa.int/ESA_Multimedia/Images/2018/04/Gaia_s_globular_clusters_and_dwarf_galaxies

M12

Mix+ (2026)

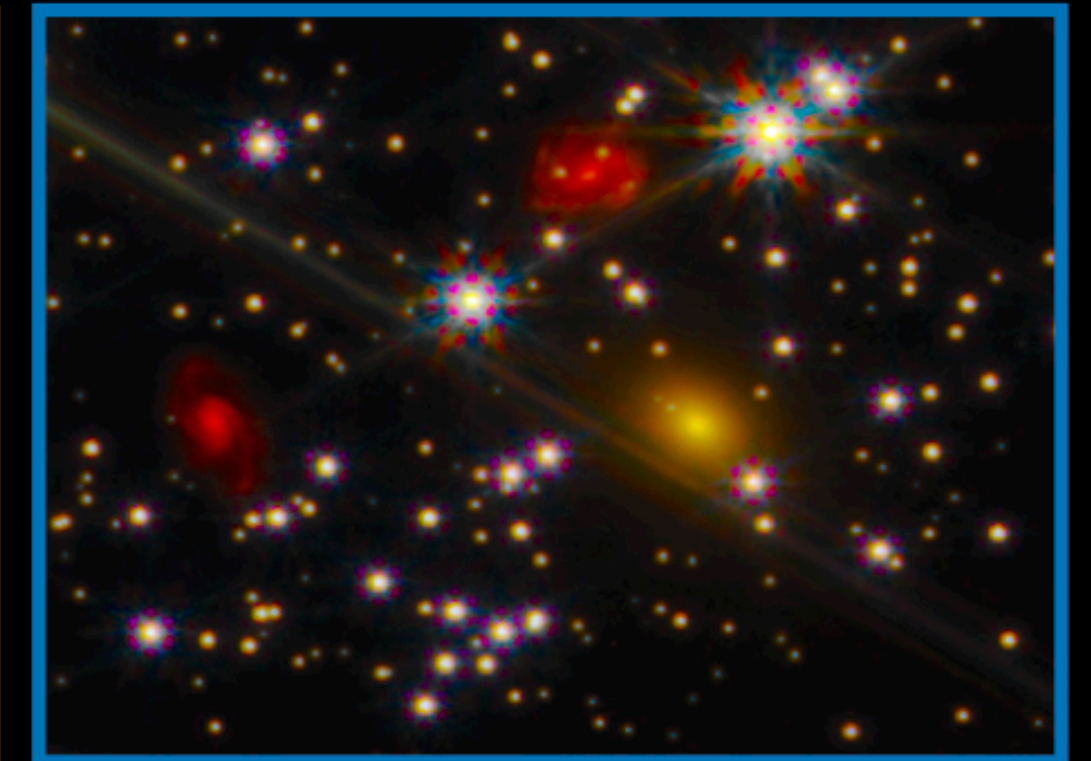
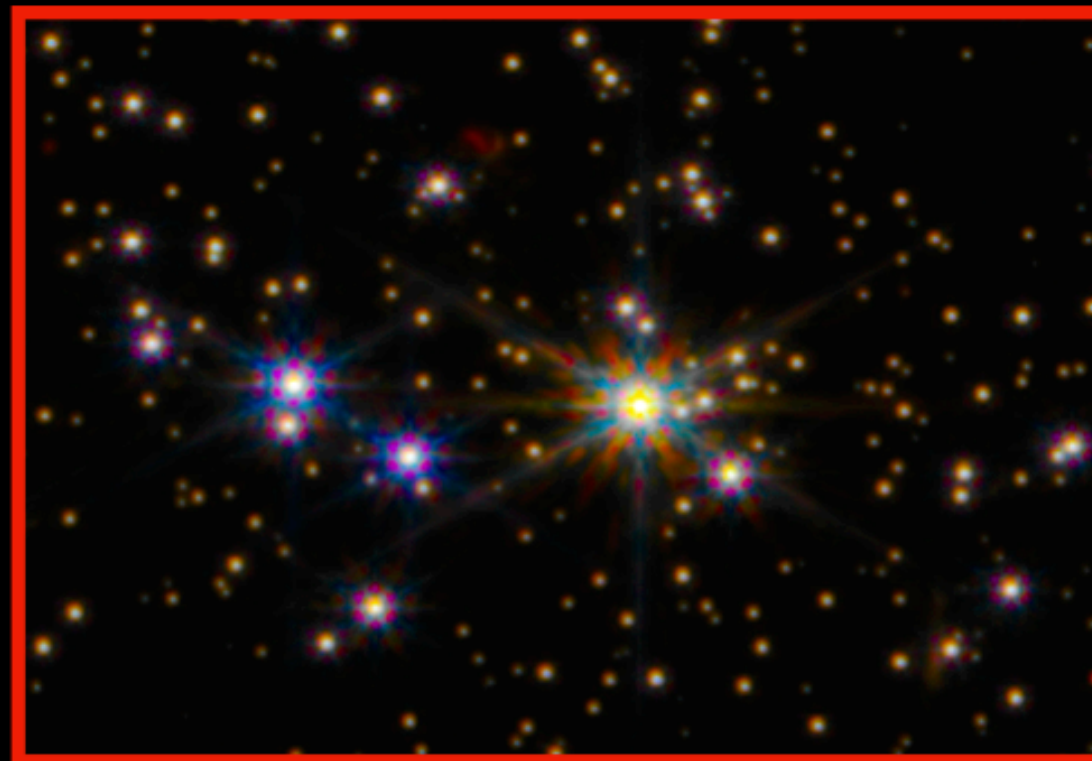
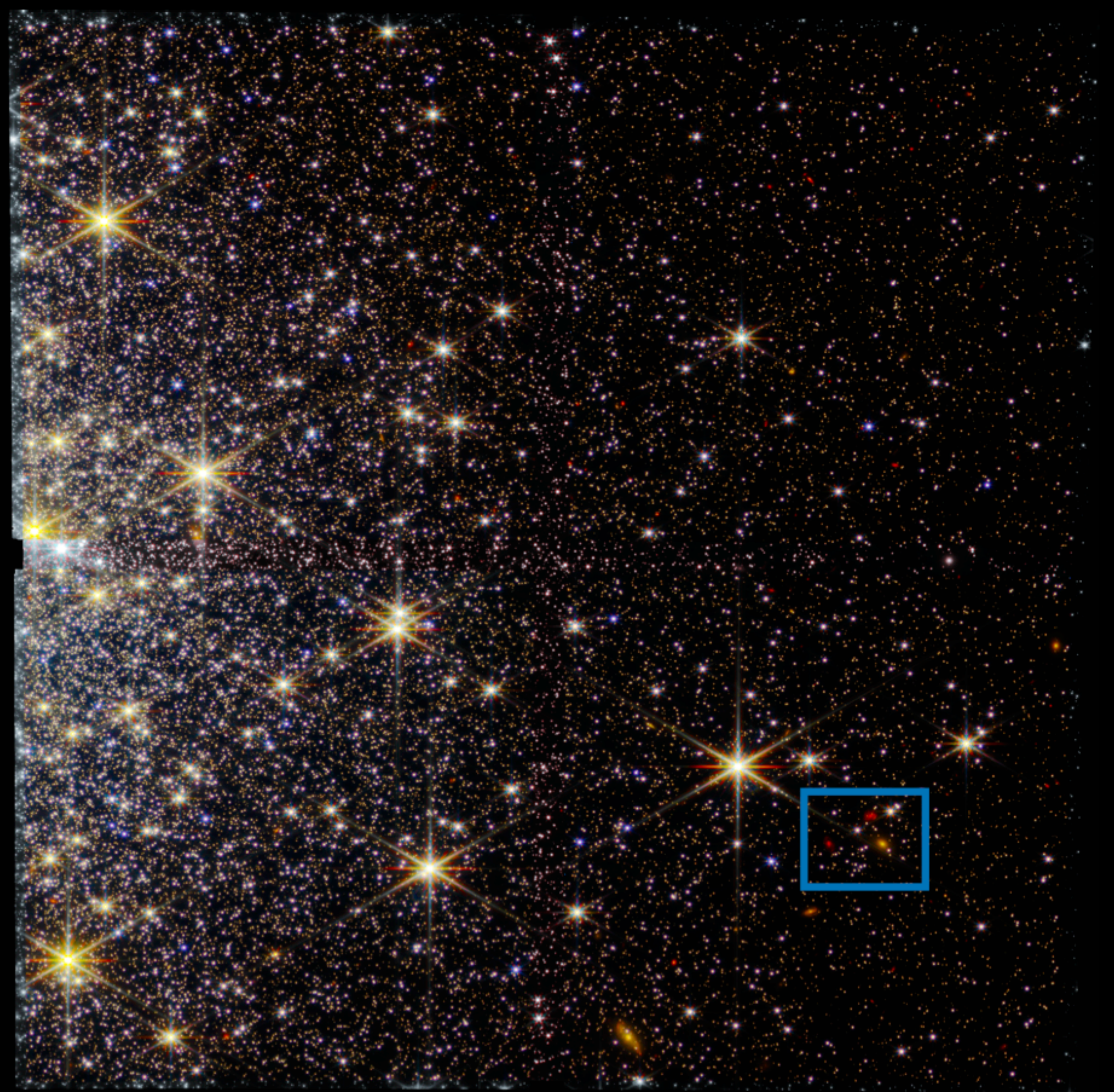


HST Color image

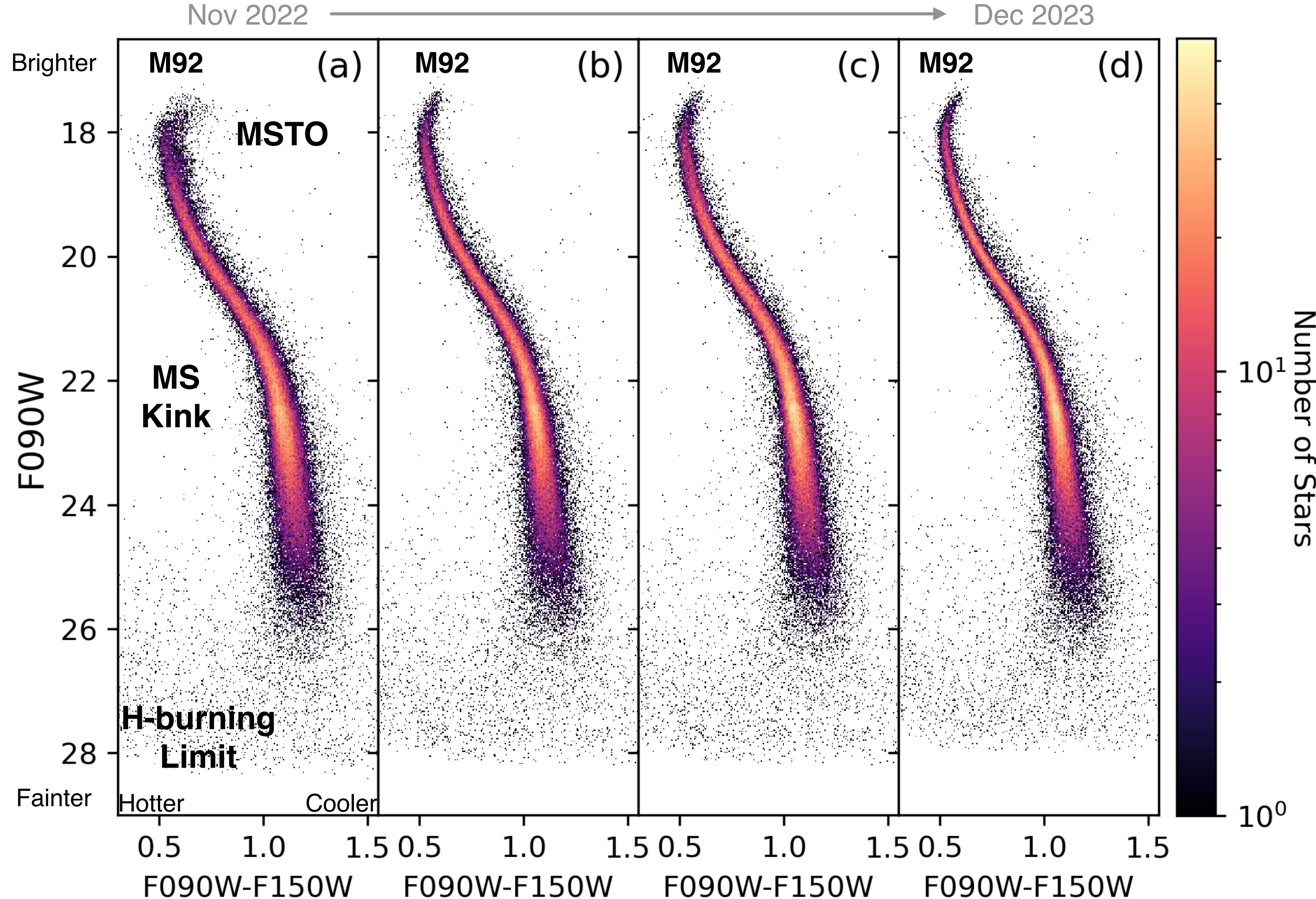


M92

Weisz+ (2023)



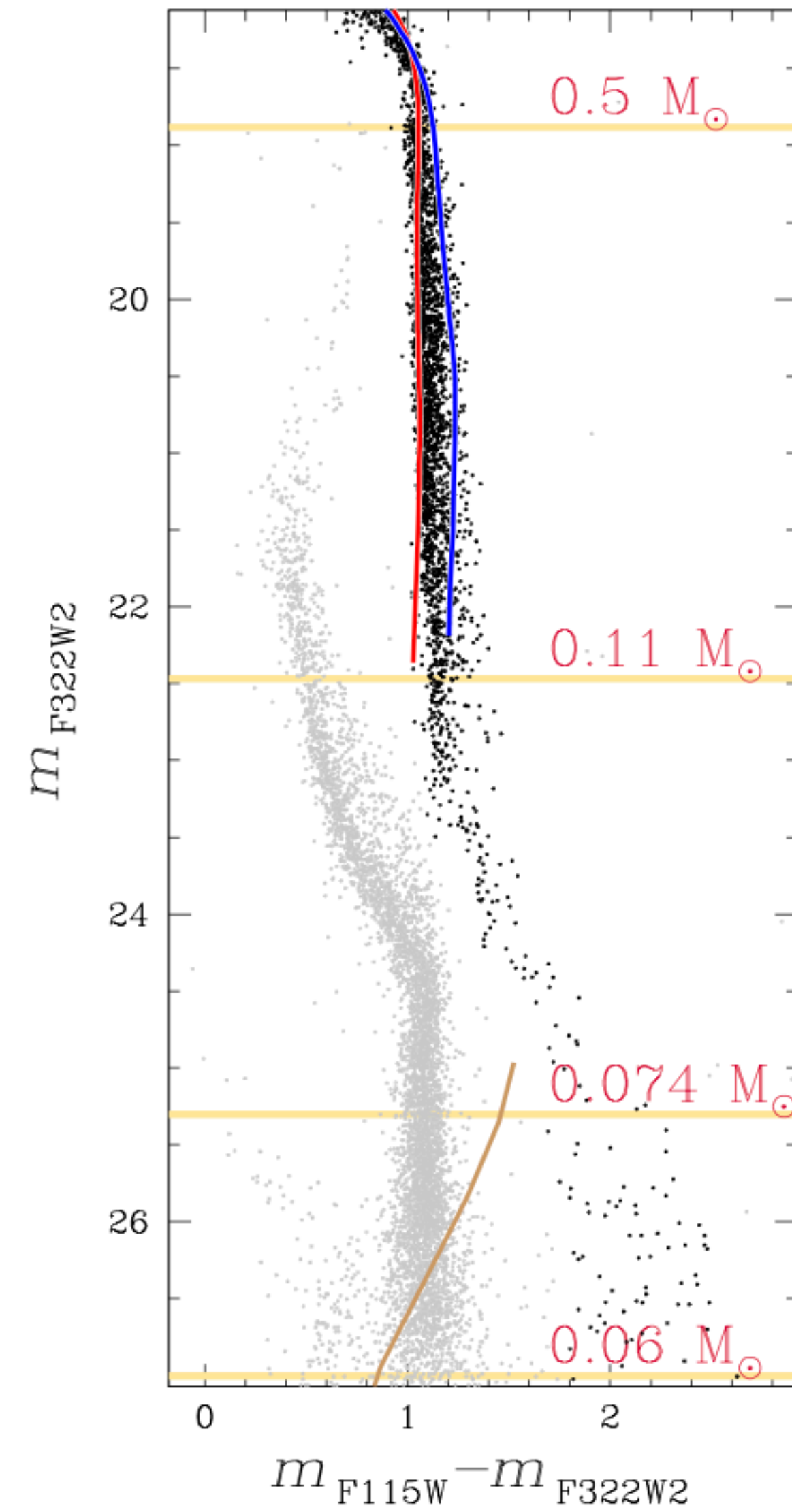
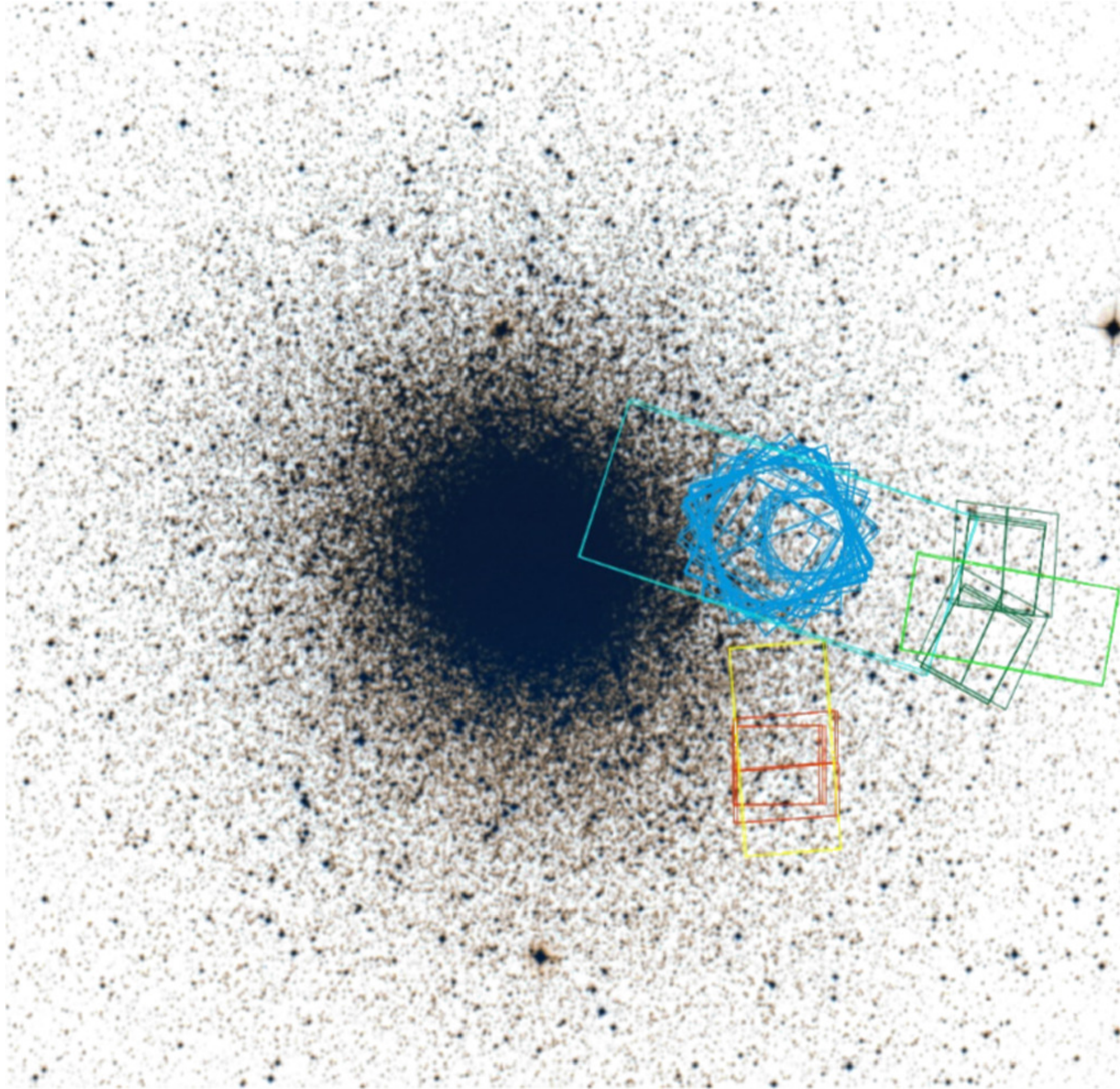
ERS: JWST-specific Crowded Field Photometry Software



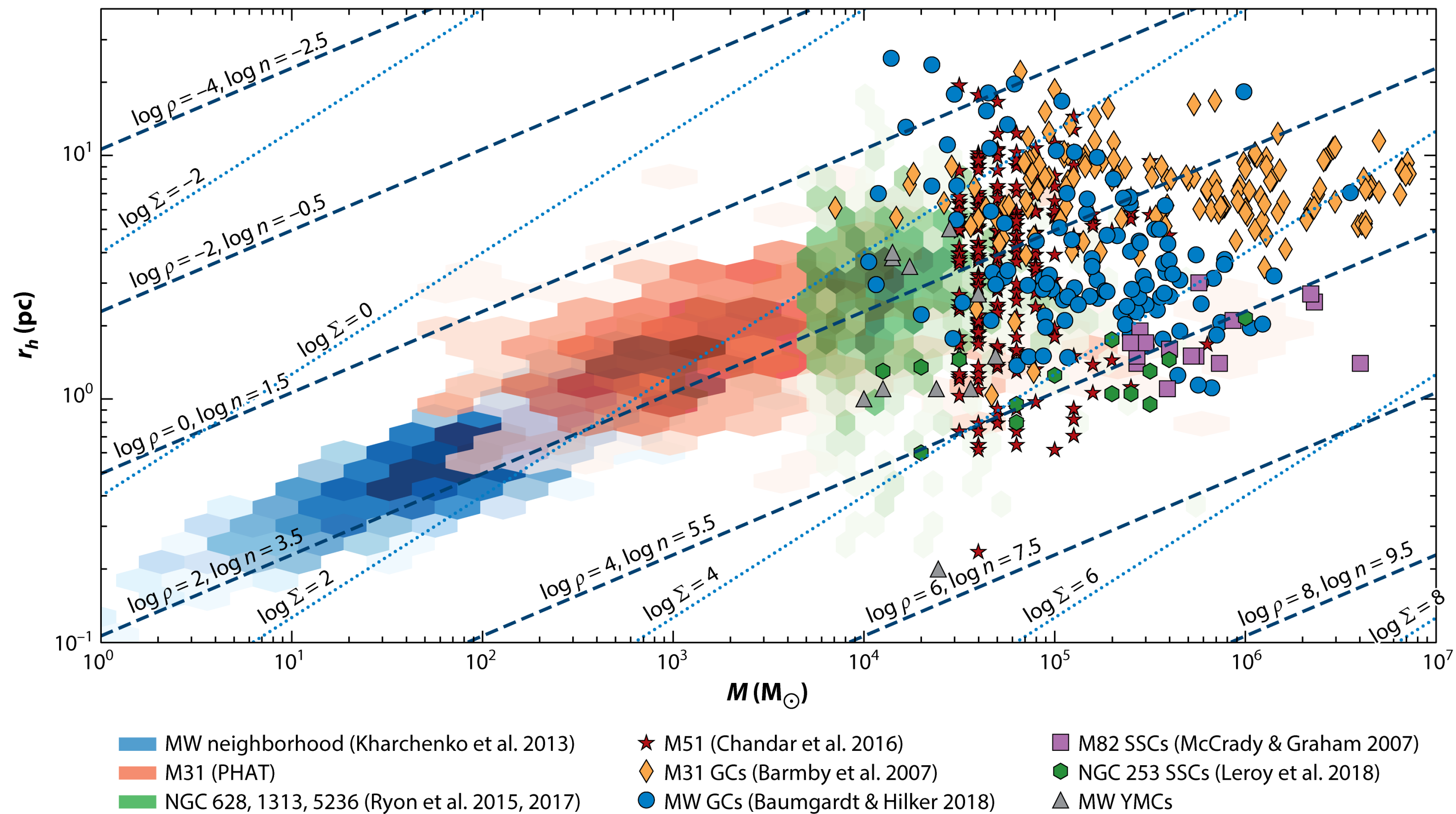
Dec 2023

- Systematics in NIRCcam crowded field photometry < 0.01 mag
- Includes detailed studies of spatial, temporal, & meteor induced impacts on photometry
- Main uncertainty: ~ 0.02 mag chip-to-chip variations in NIRCcam calibration

47 Tuc as seen with JWST



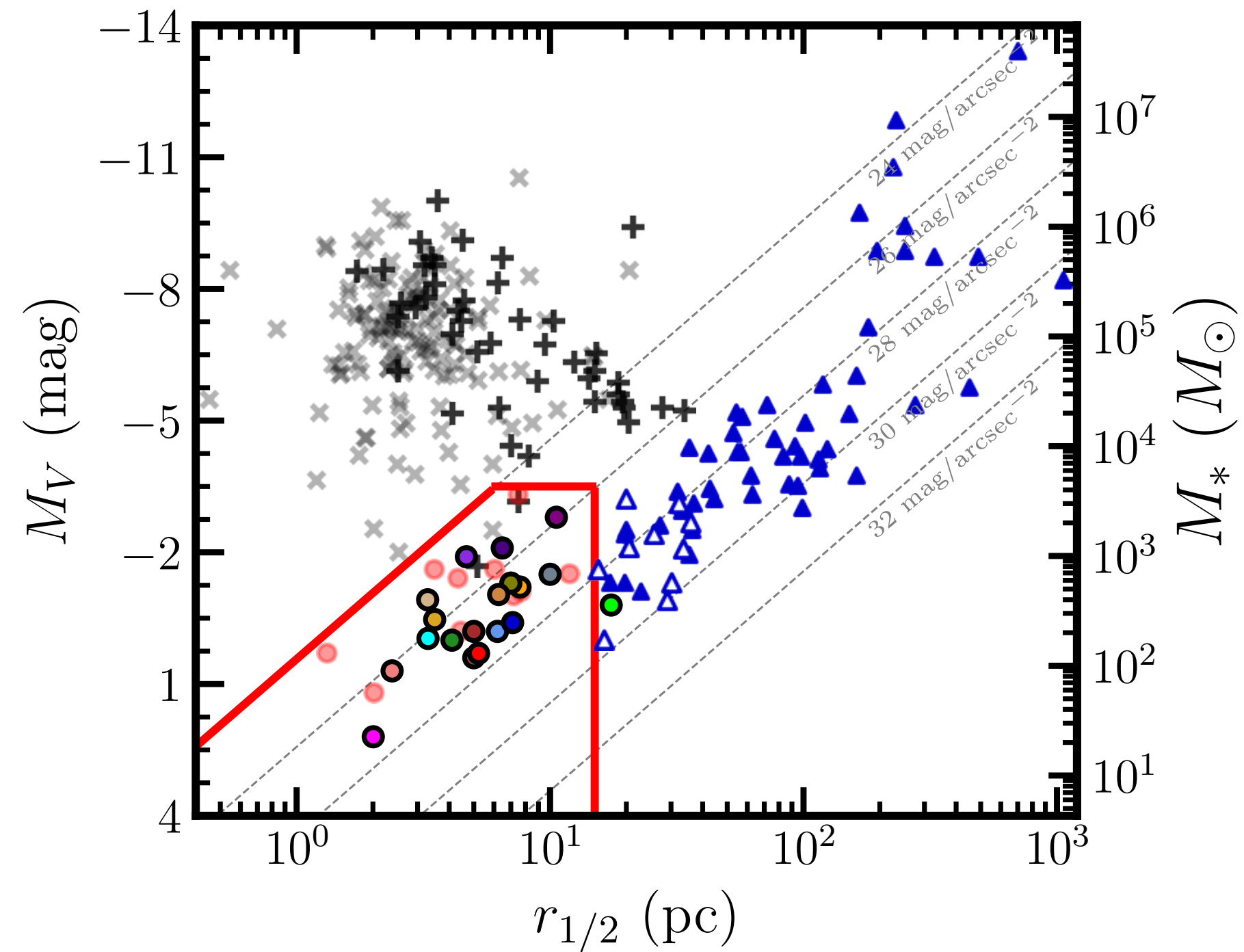
Star Cluster Mass-Size Relationship



 Krumholz MR, et al. 2019.
Annu. Rev. Astron. Astrophys. 57:227–303

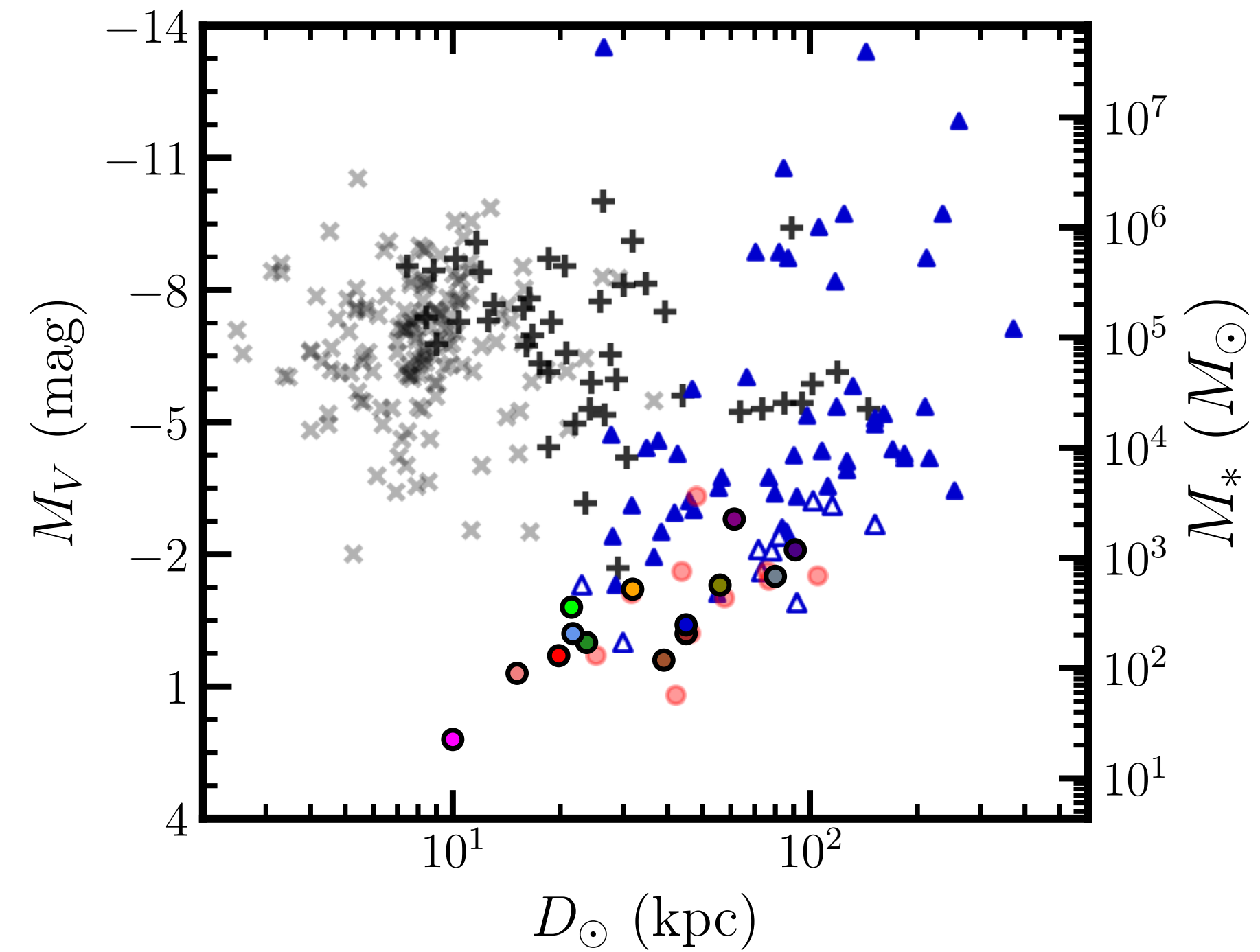
Annual Reviews

Compact Globular Clusters or Dwarf Galaxies?



Ultra-Faint Compact Satellite Spectroscopic Sample:

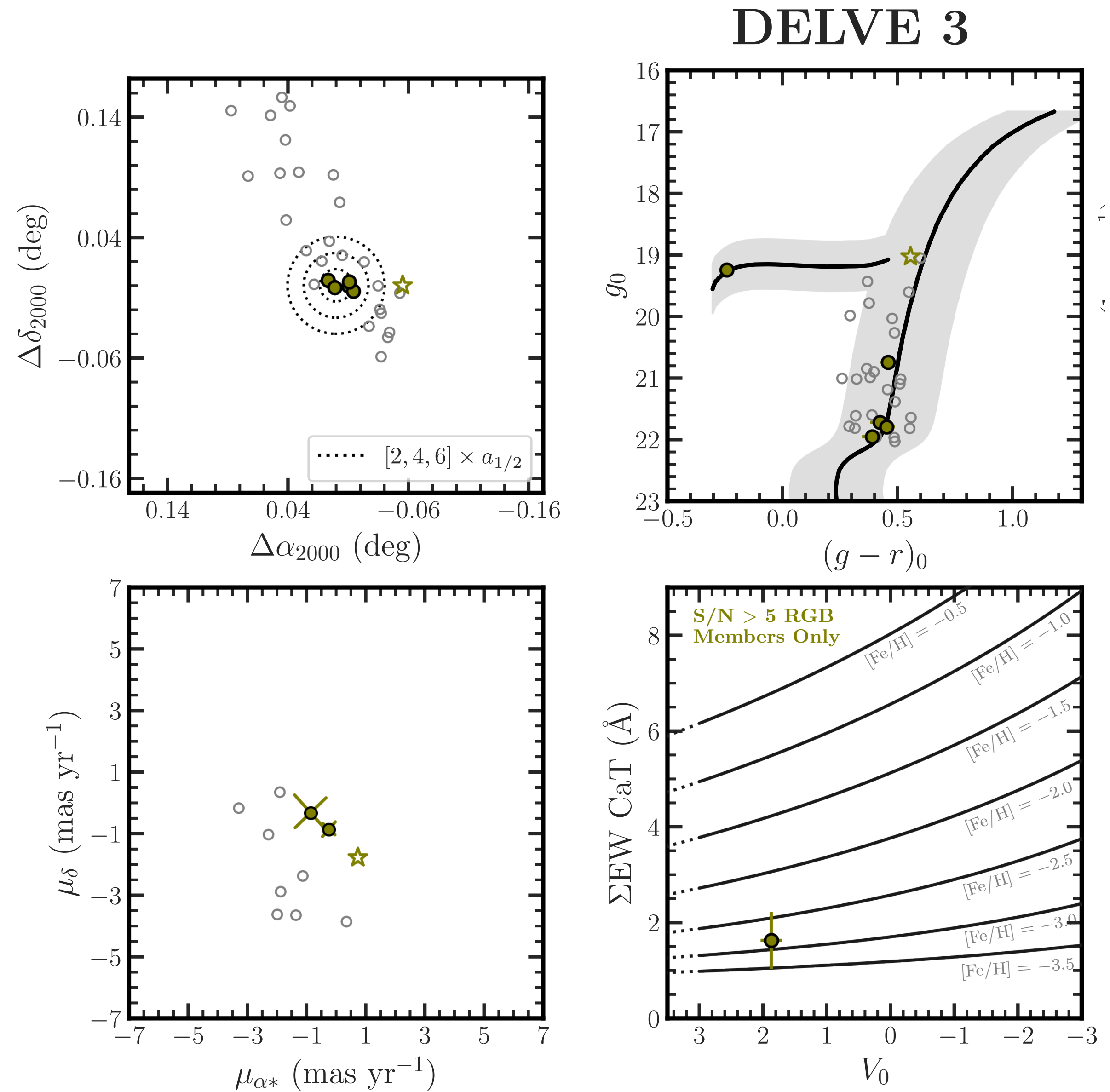
● Balbinot 1	● Draco II	● Laevens 3
● BLISS 1	● Eridanus III	● Munoz 1
● DELVE 1	● Kim 1	● PS1 1
● DELVE 3	● Kim 3	● Segue 3
● DELVE 4	● Koposov 1	● UMaIII/U1
● DELVE 5	● Koposov 2	● YMCA-1
● DELVE 6		

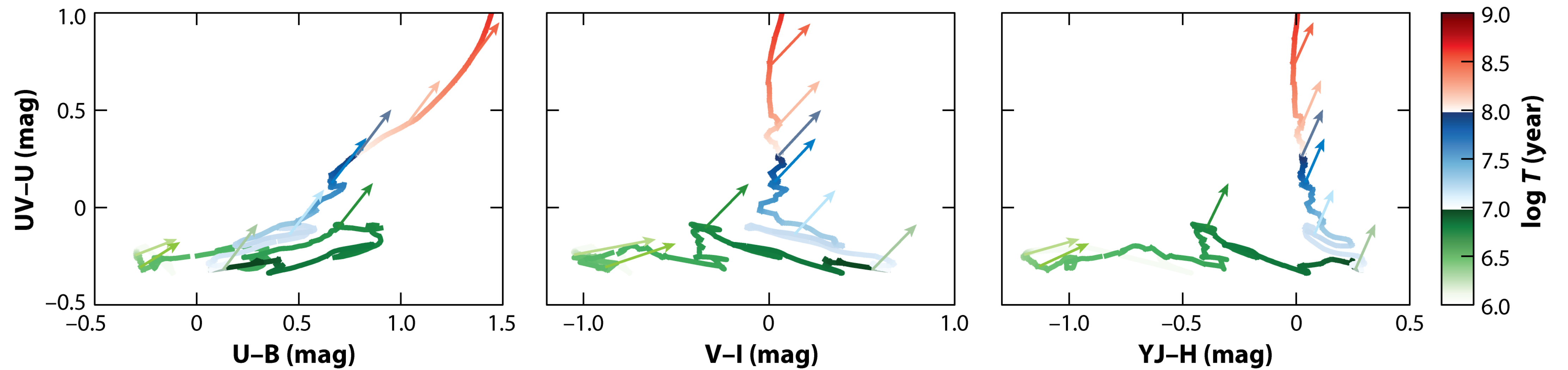


Other Milky Way Satellites:

- Ultra-Faint Compact Satellites (not in spec. sample)
- + Halo Globular Clusters ($|Z| > 5$ kpc)
- × Disk/Bulge Globular Clusters ($|Z| < 5$ kpc)
- ▲ Dwarf Galaxies ($r_{1/2} > 40$ pc or $\sigma_v \gg 0$ km s⁻¹ or $\sigma_{[\text{Fe}/\text{H}]} \gg 0$)
- △ Candidate Dwarf Galaxies

Compact Globular Clusters or Dwarf Galaxies?





AR Krumholz MR, et al. 2019.
Annu. Rev. Astron. Astrophys. 57:227–303

Annual Reviews